Introspective Imitation Dynamics and the Ceiling of Cooperation in Extrinsic Update Rules

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Abstract

Evolutionary game theory provides a framework by which to study the emergence of cooperation in a population of self-interested actors. In such a framework, players' decisions on whether or not to cooperate evolve according to decision rules called population dynamics. However, often games are studied under the assumption that all individuals play under the same conditions. As such, many common choices of update rule are not well suited for a heterogeneous population. In this paper, we categorise and compare four different population dynamics as "extrinsic", where players learn by looking outward at the other players in the population, and "intrinsic", where players look inwardly at their own attributes or potential payoffs. We show that extrinsic population dynamics have a ceiling on the rate of cooperation which can be exceeded by intrinsic population dynamics, and demonstrate this using the public goods game with heterogeneous contributions. We then introduce "introspective imitation dynamics", a novel population dynamic which combines both fitness-proportional extrinsic selection and self-referential intrinsic update steps, which overcomes the inherent limitations of the other four population dynamics.

1 Introduction

Reducing global carbon emissions is one of the defining collective action problems of our time. Countries differ substantially in their economic capacity, historical emissions, and the costs they face when decarbonising, yet they share a mutual interest in the global public good of a stable climate. However, they face the temptation to free-ride, accepting the benefits of the public good without contributing to the cost[30]. International agreements have attempted to encourage a reduction in global emissions, for example the Kyoto protocol[1] and Paris agreement[4, 16]. A recent example is the European Union's Carbon Border Adjustment Mechanism [8], which requires importers to purchase certificates reflecting the carbon content of goods produced under weaker environmental regulation, represents a real-world instance of this idea.

This structure maps onto a *public goods game* [2, 13, 15, 14, 39]: each actor decides whether to incur a private cost to contribute to a shared resource whose total value is multiplied by some factor r and distributed equally. The temptation to free-ride means that purely selfish reasoning leads to a tragedy of the commons [12, 47], even when collective cooperation would benefit everyone. Both theory [37] and controlled experiments [28, 27] show that framing cooperation as a collective-risk problem, where failure to meet a cooperation threshold incurs a shared loss, can substantially raise cooperation rates. The challenge of overcoming free-riding is not merely theoretical. Nordhaus [30] proposed voluntary *climate clubs* in which member nations adopt a common carbon price and levy tariffs on non-members, turning the international climate problem into a coordination game with enforcement.

Evolutionary game theory offers a framework for studying how cooperation can emerge among self-interested actors without assuming rational foresight [17, 31]. In this approach, strategies that perform well spread through a population via one of several update rules, while poorly performing strategies decline. The conditions under which cooperative strategies can invade and stabilise have been extensively studied in two-player settings such as the iterated prisoner's dilemma [21, 3] and the snowdrift game [46]. The public goods game generalises these pairwise dilemmas to N players [13], making it well suited to studying multi-lateral cooperation such as climate agreements. In finite populations, evolutionary outcomes are stochastic: a rare mutant strategy spreads or vanishes with a probability that depends on the population size, payoff structure, and update rule [33, 45].

A feature of real-world collective action that standard models typically set aside is that actors are not identical. Players differ in wealth, productive capacity, and the costs they face, and this inequality can reshape the dynamics of cooperation [15]. In spatial and graphical settings, where players sit on a network and participate simultaneously in multiple games with their neighbours [32, 42, 34], heterogeneity generally promotes cooperation. In [38] it is shown that social diversity in endowments promotes the emergence of cooperation in the public goods game directly: heterogeneous populations outperform their homogeneous counterparts over a wide range of parameters. This echoes earlier work showing that scale-free network topology, a form of structural heterogeneity, provides a unifying framework for the emergence of cooperation [36]. When contributions depend on dynamically accumulated reputation [26], high-reputation defectors receive reduced transfers, weakening defection's advantage. When heterogeneity arises from differences in network degree so that players allocate contributions according to the strength of their social ties [23], the payoff structure penalises high-degree defectors and protects cooperating clusters from invasion. The characteristic spatial mechanism is the formation of cooperating clusters that are more profitable than their defecting neighbours; provided r is sufficiently large, these clusters resist invasion and eventually spread. Heterogeneity does not, however, unconditionally favour cooperation. Flores et al. [9] show that when the contribution level is itself a strategic variable rather than a fixed attribute, low-value contributions act as a form of defection against the most collectively beneficial strategies, so that the emergence of cooperation does not guarantee the emergence of high-value cooperation.

The studies above are almost exclusively concerned with spatially structured populations, where network topology is central. Much less is known about fully heterogeneous, *well-mixed* populations, in which all individuals are distinguished by a fixed attribute (such as their contribution capacity) independently of any graph structure. Furthermore, most existing analyses rely on simulation or mean-field approximations; exact treatments of heterogeneous evolutionary dynamics remain rare. How the choice of update rule interacts with population heterogeneity is also poorly understood. The Moran process [29, 20], Fermi imitation dynamics [43], aspiration dynamics [7], and introspection dynamics [5, 6] all model strategy revision in distinct ways, and each may respond differently to player-specific payoffs. In particular, *intrinsic* dynamics, in which a player evaluates a prospective strategy change based on their payoff in comparison with their own counterfactual payoff or aspiration level, are better suited to heterogeneous populations than *extrinsic* dynamics, in which a player may copy a high-fitness neighbour without considering whether the same strategy would serve them as well.

We make the following contributions. First, we build on an established general framework for evolutionary dynamics on fully heterogeneous, well-mixed populations of N ordered individuals, in which each player may have a distinct payoff function and contribution level. Second, we formalize the notion of a purely extrinsic population dynamic, a definition encompassing both the Moran process [29, 20] and Fermi imitation dynamics [43], and establish that such dynamics are inherently constrained by an upper bound on the probability of cooperation. Third, we consider two intrinsic population dynamics, Aspiration dynamics [7] and Introspection dynamics [5, 6], show that these processes can overcome the ceiling of purely extrinsic dynamics, and discuss their weaknesses. Finally, we introduce a novel update rule, *Introspective Imitation Dynamics*, in which players select a role model proportionally to fitness (as in the Moran process) but evaluate whether to adopt that strategy by comparing their own current and counterfactual payoffs (as in introspection dynamics). This rule is well suited to heterogeneous populations, where copying a high-fitness individual is likely although irrational if the same strategy yields a worse payoff for the copying player. For each population dynamic, we will use the public goods game [39, 15, 2, 14, 13] to demonstrate our results.

The remainder of the paper is organised as follows. Section 2 introduces the population model and the public goods game. Section 3 introduces a formal definition of a *purely extrinsic population dynamic*, gives formulations of the Moran process and Fermi imitation dynamics in terms of our population model, proves that these satisfy the definition of a purely extrinsic population dynamic, and proves the ceiling on the probability of cooperation in such dynamics. We also demonstrate this in a public goods game. Section 4 gives formulations of two intrinsic population dynamics and shows that they can exceed the ceiling on the probability of cooperation. We also discuss their disadvantages, and show their results in the public goods game. Section 5 introduces *introspective imitation dynamics*, a novel population dynamic, and show that it exhibits none of the disadvantages present in the other four dynamics. We then show under which conditions introspective imitation dynamics is able to exceed the cooperation ceiling in the public goods game.

2 The General Heterogeneous Population Dynamic Model

As in [5, 6], we define a general framework for evolutionary dynamics on a fully heterogeneous, well-mixed population.

2.1 Population and State Space

The population consists of N **ordered** individuals. Each individual selects an action from a common action set $A = (a_1, a_2, \dots, a_k)$; we assume throughout that all individuals share the same action set of size k although this is not strictly required for introspection and aspiration dynamics [5, 7]. A *state* of the population is an ordered N-tuple $\mathbf{a} = (a_1, a_2, \dots, a_N) \in A^N$, and the *state space* S is the set of all such tuples. A payoff function $\pi \colon S \to \mathbb{R}^N$ assigns to each state a vector of individual payoffs; we write $\pi_i(\mathbf{a})$ for the payoff of individual i in state $\mathbf{a}$.

The population evolves as a Markov chain [40] on S. Transitions can only occur between states that differ in exactly one coordinate, that is, states $\mathbf{a}$ and $\mathbf{b}$ with Hamming distance $h(\mathbf{a}, \mathbf{b}) = 1$ [22]. We call the unique coordinate at which $\mathbf{a}$ and $\mathbf{b}$ differ the *index of difference*, denoted $I(\mathbf{a}, \mathbf{b})$ [6]. The set of states reachable from $\mathbf{a}$ in one step is $\text{Neb}(\mathbf{a}) = \{\mathbf{b} \in S : h(\mathbf{a}, \mathbf{b}) = 1\}$ [6].

We therefore obtain a transition matrix T, where $T_{\mathbf{ab}}$ denotes the probability of transitioning from a state $\mathbf{a}$ to a state $\mathbf{b}$. They take the general form:

$$T_{\mathbf{ab}} = \begin{cases} P(\mathbf{a} \to \mathbf{b}) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}), \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b}. \end{cases} \tag{1}$$

where P is the probability that player $I(\mathbf{a}, \mathbf{b})$ changes from its action type in $\mathbf{a}$ to its action type in $\mathbf{b}$. The process which determines this probability is called a *population dynamic*. We shall discuss five population dynamics in this paper, and classify them as shown in Figure 1. Purely extrinsic population dynamics only consider the success of a strategy in the current population when updating a player's action type. This models a behaviour where players look outwardly to attempt to improve their payoff. Intrinsic population dynamics model a player who looks inwardly to improve their payoff - comparing their current payoff to some value which does not depend on the payoffs of an action type in the current population. The final classification is our novel population dynamic, introspective imitation dynamics, which contains both an extrinsic step and an intrinsic step.

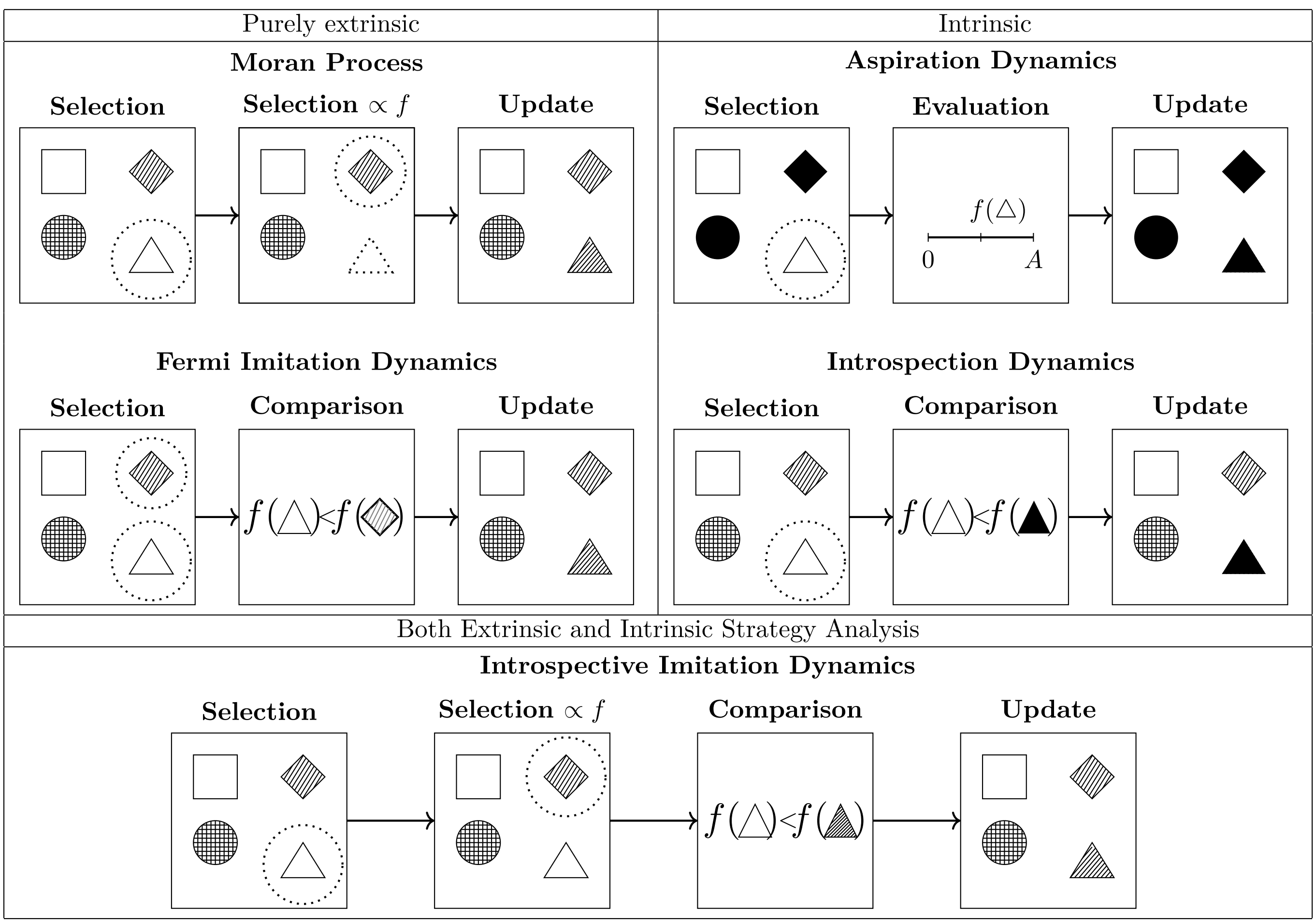


Figure 1: A comparison of different population dynamics. We assume in all cases that the triangle is chosen for replacement with a probability $\frac{1}{N}$ and the diamond is chosen with a varying probability depending on the process. In the case of Introspection dynamics, we assume that the strategy represented by the shape being filled in is chosen with probability $\frac{1}{k-1}$, and for aspiration dynamics there is only one other strategy to accept.

In the case of a game with two actions, for example cooperate and defect, the state space of the Markov chain is an N-dimensional hypercube. This is the graph whose node set consists of the 2^N N-dimensional boolean vectors, and which has edges between edges if and only if they differ in exactly one position [11]. The state space of an evolutionary game with two actions can be modelled in this way by denoting the strategies 0 and 1. The graphical representation of a birth-death process differs from a hypercube in exactly two ways. First, a population may remain the same at a given step, and thus each vertex contains a self-loop. Additionally, the probability of transitioning from a state $\mathbf{a}$ to a state $\mathbf{b}$ is not necessarily the same as the probability of transitioning from $\mathbf{b}$ to $\mathbf{a}$. Thus, instead of one edge between two states, we have two directed edges, one in each direction. In Figure 2b and 2c we see the graph of a 3 and 4 dimensional hypercubes.

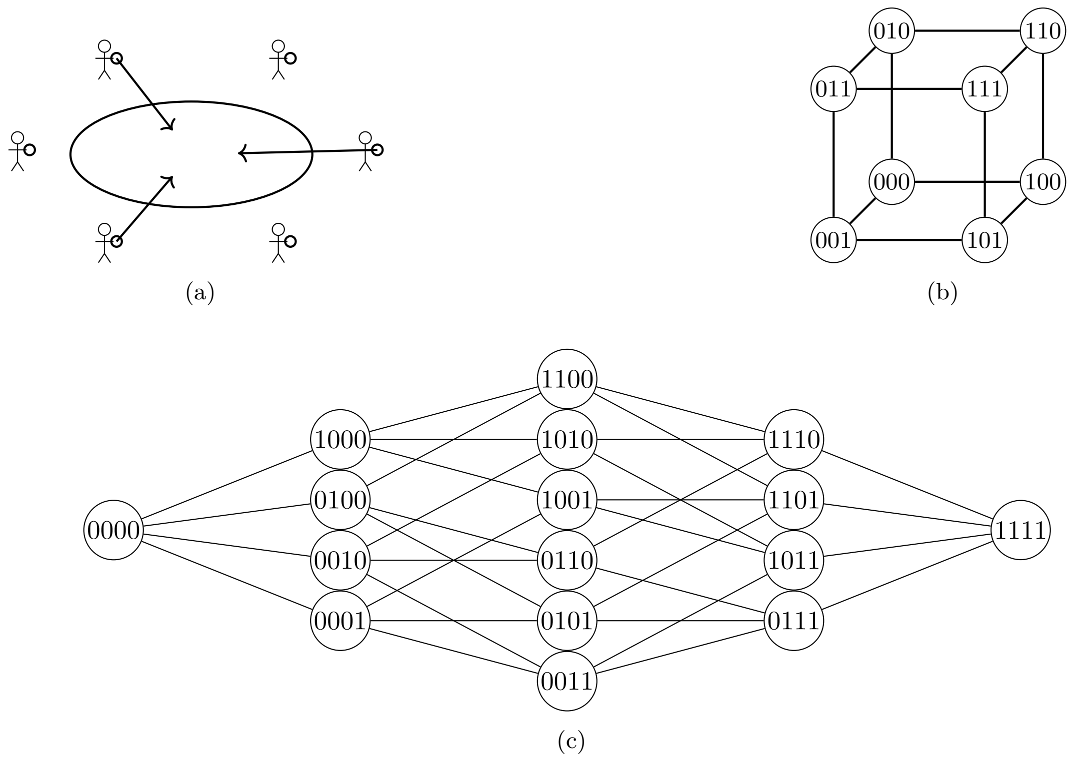


Figure 2: (a): The public goods game with 6 players. Here 3 players contribute. (b): The hypercube of dimension 3. This is closely related to the graph of a 3-player population dynamic. (c): The hypercube of dimension 4. This is closely related to the graph of a 4-player population dynamic.

2.2 Mutation

Three of the dynamics that we discuss in this paper correspond to absorbing Markov chains, where the measurable outcome is the probability of absorption into a given absorbing state. The remaining two dynamics are ergodic chains, for which we can compute the long-run abundance of a given strategy in the population. To enable meaningful comparison across all models, we introduce mutation. Mutation ensures that all processes become ergodic, while also introducing a controlled amount of noise: at any moment, individuals may switch to a different strategy [10].

Given an original process and two states $\mathbf{a}$ and $\mathbf{b}$, the probability of the original process *with mutation* transitioning from $\mathbf{a}$ to $\mathbf{b}$ is given by:

$$P(\mathbf{a} \to \mathbf{b}) = P(\text{pick individual } I(\mathbf{a},\mathbf{b})) \cdot \left(P(\mathbf{a} \to \mathbf{b}) \cdot P(\text{no mutation}) + P(\mathbf{a}_{I(\mathbf{a},\mathbf{b})} \text{ mutates} \to \mathbf{b}_{I(\mathbf{a},\mathbf{b})})\right)$$

We assume a mutation parameter $\mu \in [0,1]^{N\times k}_{\mathbb{R}}$, where μ_{ij} denotes the probability that individual i adopts strategy j via mutation. For a focal individual i, the probability that no mutation occurs is $1 - \sum_{j=1}^{k} \mu_{ij}$. Let T be the transition matrix for a population dynamic and $T_{\mathbf{ab}}$ be the transition probability from a state $\mathbf{a}$ to a state $\mathbf{b}$. Then the transition matrix $T^{(\mu)}$ for the process with mutation is given by:

$$T^{(\mu)}_{\mathbf{ab}} = \begin{cases} T_{\mathbf{ab}}\left(1-\sum_{j=1}^{k}\mu_{I(\mathbf{a},\mathbf{b}),j}\right) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),\mathbf{b}_{I(\mathbf{a},\mathbf{b})}}}{N} & \text{if } \mathbf{b}\in \text{Neb}(\mathbf{a}),\\ 0 & \text{if } \mathbf{b}\notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a}\neq\mathbf{b},\\ 1-\sum_{\mathbf{b}\in S\setminus\{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a}=\mathbf{b}. \end{cases} \tag{2}$$

Note that if $\sum_{j=1}^{k}\mu_{ij} = 1$, mutation fully determines the behavior of individual i. This construction ensures that all processes are ergodic, rather than absorbing. Consequently, we no longer obtain absorption matrices for models with mutation; instead, we derive a unique steady state that is independent of the initial state. All processes in this paper are considered under the effect of action-invariant mutation, where $\mu_{ik} = \mu_{il}$ for all action types k, l. This ensures that mutation does not cause an inherent bias towards a given action type.

2.3 The Public Goods Game

We apply the dynamics above to the public goods game [39, 15, 2, 14, 13] to illustrate the results in the following sections. In this game, individuals choose whether or not to contribute to a common resource pool. The total contribution is multiplied by a factor $r > 1$ and distributed equally among all N players. The model captures the temptation to free-ride [12]: a player who withholds their contribution still receives an equal share of the multiplied pool, so individual incentives push against the collectively optimal outcome of universal cooperation.

We can model a public goods game by providing the following payoff function:

$$\pi_i(\mathbf{a}) = \frac{r\sum_{j=1}^{N}\alpha_j}{N} - \alpha_i \tag{3}$$

Where α_j is the contribution by individual j. This is a similar heterogeneous public good game as the one described in [15] although in that model the return r is also a heterogeneous property of each player, whereas we only consider heterogeneity in contributions.

We measure the abundance of cooperation that emerges in the population, and for this purpose we define p_C to be the *probability of cooperation*. This represents the mean proportion of cooperators at a given step once an evolutionary game reaches its steady state, and can be calculated by the value $\frac{\sum_{\mathbf{a}\in S}\left(\pi_{\mathbf{a}}\sum_{i=1}^{N}\mathbf{a}_i\right)}{N}$, where $\mathbf{a}_i$ takes the value 1 if player i cooperates in state $\mathbf{a}$, and 0 if not, and $\pi_{\mathbf{a}}$ is the fraction of time spent in state $\mathbf{a}$ in the steady state of the process.

It can be seen that full contribution is the Pareto optimal state, however if $r < N$, full defection forms the Nash equilibrium for the one shot game. When $r > N$, i.e the value of the public good is multiplied by a greater amount than the number of players, each player will profit from their own contribution, as we have the value $\Delta(\pi_i) = \pi_C(\mathbf{a}) - \pi_D(\mathbf{b}) = \alpha_i(\frac{r}{N} - 1) > 0$ if and only if $r > N$. Therefore, in the case $r > N$, the Nash equilibrium of the one-shot game is full contribution. However, it is important to note that in a given mixed state in the evolutionary model of cooperation in the public goods game, a defector will have a higher payoff than a cooperator, no matter the value of r. This is because all players receive the same return from the public good, whilst only cooperators incur a cost $-\alpha_i$

To consider the distribution of the probability of cooperation in a public goods game, a large scale parameter sweep is carried out. We consider a "linear" population, in which a player in position i contributes $\frac{2Mi}{N(N+1)}$, where M is the total amount contributed when all players play C. In such a case, the payoff of all players is $\pi_i = \frac{rM}{N} - \alpha_i$. Supplementary Information Figure 1 shows the combinations of parameters considered across each of the population dynamics. To the authors knowledge this is the largest such parameter sweep. It is done to the highest standards of reproducibility [48, 35, 49, 41, 44] and all data has been archived and is a further contribution of the work. For the purpose of this research, the authors developed a software library `ludics` (written in python) which is installable. The archive of the software is at [?] and the documentation can be found at `https://hefos.github.io/ludics/`. The full development of the software can be found at `https://github.com/hefos/ludics`.

3 Extrinsic population dynamics

We begin by defining a *purely extrinsic population dynamic*. Intuitively, a purely extrinsic process revises strategies through payoff comparison between members of a population.

Definition 1 (Extrinsic population dynamic)**.** *Let T be the transition matrix of a population dynamic on a state space S. We say that T defines a* purely extrinsic *process if for every $\mathbf{a}, \mathbf{b} \in S$ with $\mathbf{b} \in$ Neb$(\mathbf{a})$, $T_{\mathbf{ab}}$ satisfies:*

1. *$T_{\mathbf{ab}}$ is a function of payoffs evaluated in $\mathbf{a}$ and in no other state.*

2. *Let $\mathcal{R}$ denote the set of players j such that $j \neq I(\mathbf{a},\mathbf{b})$ and $\mathbf{a}_j = \mathbf{b}_{I(\mathbf{a},\mathbf{b})}$. Then $T_{\mathbf{ab}}$ is non-decreasing in the payoffs $\pi_j(\mathbf{a})$ for all $j \in \mathcal{R}$ (and strictly increasing in at least one such j if $\mathcal{R} \neq \emptyset$), and non-increasing in the payoffs $\pi_j(\mathbf{a})$ for all $j \notin \mathcal{R}$.*

3. *$T_{\mathbf{ab}}$ contains no term dependent on $a_{I(\mathbf{a},\mathbf{b})}$ or $b_{I(\mathbf{a},\mathbf{b})}$, except in the payoff function $\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a})$, mutation parameters μ_{ij}, and in the definition of the set $\mathcal{R}$.*

4. *If all players share the same payoff in state $\mathbf{a}$ (that is, there exists $v \in \mathbb{R}$ such that $\pi_i(\mathbf{a}) = v$ for every player i), then $T_{\mathbf{ab}}$ is independent of the common value v.*

5. *Let T^{nd} denote T with all payoffs set to a common value, well-defined by (4). Order $\{C, D\}^N$ componentwise with $D < C$. For every $\mathbf{a} \leq \mathbf{a}'$, there exist random variables $\mathbf{X}, \mathbf{X}'$ on a common probability space, with $\mathbf{X} \sim T^{\mathrm{nd}}(\mathbf{a}, \cdot)$ and $\mathbf{X}' \sim T^{\mathrm{nd}}(\mathbf{a}', \cdot)$, such that $\mathbf{X} \leq \mathbf{X}'$ almost surely.*

Condition (2) captures the basic monotonicity: a higher payoff for a role model makes that role model more likely to be copied, and a higher payoff for a non-role-model pulls the other way. Condition (3) rules out an intrinsic preference for either action. Conditions (4) and (5) work as a pair, ruling out hidden biases that survive when no payoff comparison is available.

Condition (4) captures payoff-scale neutrality: when all players earn the same payoff, the absolute level of that common value cannot influence $T_{\mathbf{ab}}$. Condition (5) captures configuration neutrality: under those same neutral payoffs, starting from a more cooperative profile cannot make the next state any less cooperative. Put simply: at neutral payoffs, a population that starts at least as cooperative as another remains at least as cooperative one step later. Without (5), a process could satisfy (1)–(4) and still embed a configuration-dependent pull toward defection at neutral payoffs, which is an intrinsic bias rather than truly neutral behaviour. [19] notes that birth-death processes are stochastically monotone, and we shall show that the Moran process and Fermi imitation dynamics are also stochastically monotone.

Note: Condition (3) in the above definition serves to exclude population dynamics which explicitly favour a certain action type, for example:

$$\bar{T}_{\mathbf{ab}} = \begin{cases} \delta_{I(\mathbf{a},\mathbf{b})} \cdot T_{\mathbf{ab}} & \text{if } \mathbf{b} \neq \mathbf{a}, \\ 1 - \sum_{\mathbf{c} \neq \mathbf{a}} T_{\mathbf{ac}} & \text{if } \mathbf{b} = \mathbf{a}. \end{cases}$$

where $T_{\mathbf{ab}}$ is the transition matrix of some purely extrinsic dynamic, and $\delta_{I(\mathbf{a},\mathbf{b})}$ takes the value 1 if $b_{I(\mathbf{a},\mathbf{b})} = C$ and $\frac{1}{2}$ if not. Without (3), $\bar{T}$ would define purely extrinsic population dynamic, but would be tailored to favour a particular action type. We exclude cases such as this, as they do not conform to the framework in which fitness is the single measure of the success of action types in the population, and thus of which strategy is passed on in each step of the process. While parameters such as choice intensity and selection intensity affect the transition probability $T_{\mathbf{ab}}$, they do not inherently favour any specific action type.

We now show two classical examples of purely extrinsic population dynamics

3.1 The Moran Process

The Moran process [29] is defined by the following algorithm:

1. A player is chosen to be copied with a probability proportional to their fitness in the population
2. A player is chosen with probability $\frac{1}{N}$ to change their strategy

The transition matrix which defines the Moran process for a given game is:

$$T_{\mathbf{ab}} = \begin{cases} \dfrac{\sum_{j:\, a_j = b_{I(\mathbf{a},\mathbf{b})}} f_j(\mathbf{a})}{N \sum_j f_j(\mathbf{a})} \left(1 - \sum_{j=1}^{k} \mu_{I(\mathbf{a},\mathbf{b}),j}\right) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),\mathbf{b}_{I(\mathbf{a},\mathbf{b})}}}{N} & \text{if } \mathbf{b} \in \mathrm{Neb}(\mathbf{a}), \\ 0 & \text{if } \mathbf{b} \notin \mathrm{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b}. \end{cases} \tag{4}$$

Where $f_i(\mathbf{a}) = 1 + \epsilon \pi_i(\mathbf{a})$ is the fitness of a player i in state $\mathbf{a}$ scaled with selection intensity ϵ. This ensures a positive fitness for all players in order that the probability $T_{\mathbf{ab}}$ is always well defined. The selection intensity ϵ controls the rationality with which players choose which strategy to adopt in each time step - a larger ϵ indicates that a player will choose a higher fitness strategy with a greater probability. This value of ϵ must, however, be chosen in order that $f_i(\mathbf{a})$ is strictly positive.

The Moran process is traditionally employed as a model of natural evolution [31, 29], where a player's "type" represents a mutation within a population. In each time step, rather than a given player choosing a new action type, a randomly chosen individual in the population would die and be removed from the population. Another individual would reproduce with a probability proportional to their fitness, introducing an exact copy of themselves to the population to replace the removed individual. As a result of the model's original purpose, the Moran process contains no decision step where a player chooses whether or not to accept a considered strategy, as in natural selection a player cannot choose their action type. Instead, at each time step, a player *must* accept another player's action, and we remain in the same state when a player accepts the same action as they currently play.

One implication of this is that in a heterogeneous population, worsening moves may be taken with a high probability. This is because an action that provides a high payoff for one player in a population may not provide the same return for another. In nature, an offspring is unable to choose its parents, and thus cannot choose which attributes it is born with. Furthermore, parents will choose a partner based on its success in the population, and thus traits which give a high fitness will be passed on with high probabilities, regardless of how they will perform for the offspring. However, when modelling the adoption of behaviours, these assumptions are not realistic, and may often lead to players making decisions which are harmful for their own payoff. For example, in the public goods game with heterogeneous returns, a player with a low $r_i < N$ may copy a player who contributes with a high $r_j > N$ due to their high individual payoff. However, this would be a worsening move for the focal player i, as due to their low individual rate of return they would lower their payoff.

We now show that the Moran process satisfies the conditions of Definition 1

Theorem 1 (The Moran process is a purely extrinsic process)**.** *The Moran process, as defined by Equation* (4)*, is a purely extrinsic population dynamic.*

Proof. By definition the Moran process satisfies conditions (1) and (3). It remains to be shown that $T_{\mathbf{ab}}$ is non-decreasing in $\pi_j(\mathbf{a})$ for $j \in \mathcal{R}$ (strictly increasing for at least one such j) and non-increasing in $\pi_j(\mathbf{a})$ for $j \notin \mathcal{R}$. Recall $f_i(\mathbf{a}) = 1 + \epsilon\pi_i(\mathbf{a})$ for some selection intensity $\epsilon > 0$. Then we have:

$$T_{\mathbf{ab}} = \frac{\sum_{j\in\mathcal{R}} f_j(\mathbf{a})}{N\sum_{l=1}^{N} f_l(\mathbf{a})}\left(1 - \sum_{i=1}^{k}\mu_{I(\mathbf{a},\mathbf{b}),i}\right) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),b_{I(\mathbf{a},\mathbf{b})}}}{N}$$

$$\frac{dT_{\mathbf{ab}}}{d\pi_j(\mathbf{a})} = \frac{\epsilon\sum_{x\notin\mathcal{R}} f_x(\mathbf{a})}{N\left(\sum_{l=1}^{N} f_l(\mathbf{a})\right)^2}\left(1 - \sum_{i=1}^{k}\mu_{I(\mathbf{a},\mathbf{b}),i}\right) \quad \text{for } j \in \mathcal{R}$$

For $j \in \mathcal{R}$, as ϵ is chosen such that $f_i(\mathbf{a}) > 0$ for all $i \in \mathbf{a}$, and $\mu_{i,j}$ such that $\sum_{i=1}^{k}\mu_{I(\mathbf{a},\mathbf{b}),i} < 1$, this derivative is strictly positive. For $j \notin \mathcal{R}$, the analogous computation gives $\frac{dT_{\mathbf{ab}}}{d\pi_j(\mathbf{a})} = -\frac{\epsilon\sum_{x\in\mathcal{R}} f_x(\mathbf{a})}{N(\sum_{l=1}^{N} f_l(\mathbf{a}))^2}(1 - \sum_{i=1}^{k}\mu_{I(\mathbf{a},\mathbf{b}),i}) \leq 0$, so condition (2) holds. Finally, for condition (4), setting $\pi_i(\mathbf{a}) = v$ for every player i gives $f_i(\mathbf{a}) = 1 + \epsilon v$ for every i, whence $T_{\mathbf{ab}} = \frac{|\mathcal{R}|(1+\epsilon v)}{N\cdot N(1+\epsilon v)}(1 - \sum_i \mu_{I(\mathbf{a},\mathbf{b}),i}) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),b_{I(\mathbf{a},\mathbf{b})}}}{N} = \frac{|\mathcal{R}|}{N^2}(1 - \sum_i \mu_{I(\mathbf{a},\mathbf{b}),i}) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),b_{I(\mathbf{a},\mathbf{b})}}}{N}$, which is independent of v.

For condition (5), specialise to two actions $\{C, D\}$ with $D < C$. Under T^{nd}, the dynamic samples a focal player i uniformly, a role model k uniformly from the population (since all fitnesses are equal), and a mutation outcome m from the action-invariant mutation distribution; player i then adopts a_k, or m if a mutation occurs. Couple the chains started at $\mathbf{a} \leq \mathbf{a}'$ on the common probability space generated by (i, k, m), applying the same draw to both. Coordinate i in the successor of $\mathbf{a}$ is a_k (or m); coordinate i in the successor of $\mathbf{a}'$ is a'_k (or the same m). Since $\mathbf{a} \leq \mathbf{a}'$ gives $a_k \leq a'_k$, and every other coordinate j is left unchanged with $a_j \leq a'_j$, the two successors $\mathbf{X}$ and $\mathbf{X}'$ satisfy $\mathbf{X} \leq \mathbf{X}'$ pointwise on this probability space, so condition (5) holds. □

3.2 Fermi Imitation Dynamics

The Fermi Imitation Dynamic is another method of modelling evolution in a population. It models the following process:

- Two players, i and j are selected at random. Unlike the original formulation [43], player j need not be a neighbour of player i, reflecting the well-mixed setting considered here.
- Player i copies the action type of the player j with a probability according to the Fermi function $\phi(\pi_i(\mathbf{a}) - \pi_j(\mathbf{a})) = \frac{1}{1+e^{(\beta(\pi_i(\mathbf{a})-\pi_j(\mathbf{a})))}}$

Where β is the choice intensity of the system, representing how sensitive our players' choice is to the difference between their current payoff, and the payoff which they are considering. This method has been commonly used in studies of evolutionary games, for example [26, 23, 43]. We now define our transition matrix according to the Fermi function:

$$T_{\mathbf{ab}} = \begin{cases} \frac{1}{N(N-1)} \sum_{j:\, a_j = b_{I(\mathbf{a},\mathbf{b})}} \phi\big(\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a}) - \pi_j(\mathbf{a})\big) \Big(1 - \sum_{j=1}^{k} \mu_{I(\mathbf{a},\mathbf{b}),j}\Big) + \frac{\mu_{I(\mathbf{a},\mathbf{b}), \mathbf{b}_{I(\mathbf{a},\mathbf{b})}}}{N} & \text{if } \mathbf{b} \in \mathrm{Neb}(\mathbf{a}), \\ 0 & \text{if } \mathbf{b} \notin \mathrm{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b}. \end{cases} \tag{5}$$

Unlike a Moran Process, we can see that the players are not selected with a probability proportional to their fitness. Instead, Fermi Imitation Dynamics compares the fitness after the selection of players. This gives the comparison step in which a player may choose not to adopt a new strategy if it admits a lower fitness in the current population. However, equivalently to the Moran process, in a heterogeneous population the method of adopting a new strategy based entirely on the payoff of another individual may lead to worsening moves in which players adopt strategies which currently admit a higher fitness, but which will not improve their own fitness.

We also see that due to the formulation of the logit function $\phi(\Delta(\pi))$, Fermi imitation dynamics can become invariant under certain parameters. This occurs in any game in which one action type's payoff in any given state can be expressed as another action type's payoff up to the addition of some factor γ, where γ depends only on a subset of the game's parameters. In that case, $\Delta(\pi) = \pm\gamma$, and is therefore invariant to any parameters of the game which do not influence γ. Take, for example, the public goods game. When comparing a cooperator and defector under ϕ, we have the value $\Delta(\pi) = \pm\alpha_i$, and thus we see that the parameter r has no effect on the probability of accepting a new strategy under this population dynamic.

We now show that Fermi imitation dynamics satisfies the definition of a purely extrinsic population dynamic.

Theorem 2 (Fermi imitation dynamics is a purely extrinsic population dynamic)**.** *Fermi imitation dynamics, as defined by Equation* (5)*, is a purely extrinsic population dynamic in the sense of Definition 1.*

Proof. By definition, Fermi imitation dynamics satisfies conditions (1) and (3). It remains to show that it satisfies conditions (2), (4), and (5).

$$T_{\mathbf{ab}} = \frac{1}{N(N-1)} \sum_{j \in \mathcal{R}} \phi(\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a}) - \pi_j(\mathbf{a})) \left(1 - \sum_{i=1}^{k} \mu_{I(\mathbf{a},\mathbf{b}),i}\right) + \frac{\mu_{I(\mathbf{a},\mathbf{b}), b_{I(\mathbf{a},\mathbf{b})}}}{N}$$

As ϕ is strictly decreasing, every term in the sum is strictly increasing in $\pi_j(\mathbf{a})$ for $j \in \mathcal{R}$. For $j \notin \mathcal{R}$ with $j \neq I(\mathbf{a}, \mathbf{b})$, $\pi_j(\mathbf{a})$ does not appear in the formula at all. For $j = I(\mathbf{a}, \mathbf{b})$ (when $a_{I(\mathbf{a},\mathbf{b})} \neq b_{I(\mathbf{a},\mathbf{b})}$, so $I \notin \mathcal{R}$), each term is strictly decreasing in $\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a})$ because ϕ is. Thus, Fermi imitation dynamics satisfies condition (2). Finally, condition (4) holds: if $\pi_i(\mathbf{a}) = v$ for every player i, then every payoff difference appearing in the Fermi formula is zero, so $\phi(\pi_I(\mathbf{a}) - \pi_j(\mathbf{a})) = \phi(0) = \frac{1}{2}$ in each summand, and $T_{\mathbf{ab}} = \frac{|\mathcal{R}|}{2N(N-1)}(1 - \sum_i \mu_{I(\mathbf{a},\mathbf{b}),i}) + \frac{\mu_{I(\mathbf{a},\mathbf{b}), b_{I(\mathbf{a},\mathbf{b})}}}{N}$, independent of v.

For condition (5), specialise to two actions $\{C, D\}$ with $D < C$. Under T^{nd}, the dynamic samples a focal player i uniformly, a role model k uniformly from the remaining $N - 1$ players, an independent Bernoulli outcome with success probability $\phi(0) = \frac{1}{2}$, and a mutation outcome m from the action-invariant mutation distribution; player i adopts a_k on a success (or m if a mutation occurs) and otherwise keeps a_i. Couple the chains started at $\mathbf{a} \leq \mathbf{a}'$ on the common probability space generated by $(i, k, \mathrm{Bernoulli}, m)$, applying the same draw to both. On a mutation step, both successors set coordinate i to the same m. On a successful imitation step, they set coordinate i to a_k and a'_k respectively, with $a_k \leq a'_k$. On a no-update step, coordinate i retains $a_i \leq a'_i$. Every other coordinate is unchanged, so the two successors $\mathbf{X}$ and $\mathbf{X}'$ satisfy $\mathbf{X} \leq \mathbf{X}'$ pointwise on this probability space, and condition (5) holds. □

3.3 The Cooperation Ceiling under Purely Extrinsic Population Dynamics

We will now show that a there is a theoretic limit on the probability of cooperation p_C for any purely extrinsic population dynamic. Before stating the theorem we record a comparison result for Markov chains on partially ordered spaces. The result is classical and we claim no novelty; we include a short proof for the reader's convenience, since its use in the proof of the theorem below may not be familiar to the reader.

Lemma 1. *Let P and Q be irreducible, aperiodic transition matrices on a finite partially ordered set $(S, \leq)$, with stationary distributions π^P and π^Q. Suppose that*

(i) $P(\mathbf{a}, \cdot) \preceq_{st} Q(\mathbf{a}, \cdot)$ for every $\mathbf{a} \in S$; and

(ii) Q is stochastically monotone: $\mathbf{a} \leq \mathbf{a}'$ implies $Q(\mathbf{a}, \cdot) \preceq_{st} Q(\mathbf{a}', \cdot)$.

Then $\pi^P \preceq_{st} \pi^Q$.

Proof. Kamae, Krengel, and O'Brien [18] prove that whenever two probability measures μ_1, μ_2 on a partially ordered Polish space satisfy $\mu_1 \preceq_{st} \mu_2$, there exist random variables $X_1 \sim \mu_1$ and $X_2 \sim \mu_2$ defined on a common probability space with $X_1 \leq X_2$ almost surely (a *monotone coupling*).

Using this, we construct a coupling of the two chains by induction on t. Start with any $X_0^P, X_0^Q \in S$ satisfying $X_0^P \leq X_0^Q$. Given $X_t^P \leq X_t^Q$, hypothesis (ii) gives $Q(X_t^P, \cdot) \preceq_{st} Q(X_t^Q, \cdot)$, which combined with (i) at $\mathbf{a} = X_t^P$ yields $P(X_t^P, \cdot) \preceq_{st} Q(X_t^Q, \cdot)$. Applying Kamae–Krengel–O'Brien to this pair of measures produces (X_{t+1}^P, X_{t+1}^Q) with the required marginals and $X_{t+1}^P \leq X_{t+1}^Q$ almost surely. By induction, $X_t^P \leq X_t^Q$ for every t. Since both chains are ergodic, their marginal laws converge to π^P and π^Q, and passing to the limit gives $\pi^P \preceq_{st} \pi^Q$. □

Theorem 3 (Extrinsic Ceiling)**.** *Let T define a purely extrinsic process on an N-player game with 2 actions, C and D, such that in any given state $\mathbf{a}$, $\min_{j:a_j=D} f_j(\mathbf{a}) > \max_{i:a_i=C} f_i(\mathbf{a})$ (that is, in any given state, any defector has greater fitness than any cooperator). Suppose further that the system exhibits action-invariant mutation $\mu_{iC} = \mu_{iD}$ for all players $i \in \mathbf{a}$.*

Then, $p_C \leq \frac{1}{2}$

Proof. Equip $S = \{C, D\}^N$ with the componentwise partial order, declaring $D < C$. For a transition matrix P, write $P(\mathbf{a}, \cdot) \preceq_{st} P(\mathbf{a}', \cdot)$ to mean that the one-step distribution from $\mathbf{a}$ under P is stochastically dominated by the one-step distribution from $\mathbf{a}'$ under P in this order. Let $T^{\text{nd}}_{\mathbf{ab}}$ denote the value of $T_{\mathbf{ab}}$ evaluated at $\pi_i(\mathbf{a}) = 0$ for every i; by condition (4), $T^{\text{nd}}_{\mathbf{ab}}$ is equally the value of $T_{\mathbf{ab}}$ under any other choice of common payoff.

Step 1: $T(\mathbf{a}, \cdot) \preceq_{st} T^{\text{nd}}(\mathbf{a}, \cdot)$ for every $\mathbf{a} \in S$.

Set $v(\mathbf{a}) := \min_{a_j=D} \pi_j(\mathbf{a})$. In every mixed state $\mathbf{a}$, the payoff-dominance hypothesis implies $\pi_i(\mathbf{a}) < v(\mathbf{a})$ for every cooperator i and $\pi_j(\mathbf{a}) \geq v(\mathbf{a})$ for every defector j. Consider a C-gaining transition $\mathbf{a} \to \mathbf{b}$, so $b_{I(\mathbf{a},\mathbf{b})} = C$ and $\mathcal{R}$ is the set of current cooperators. Replacing every $\pi_i(\mathbf{a})$ with $v(\mathbf{a})$ raises all role-model payoffs and lowers all non-role-model payoffs, so by condition (2),

$$T_{\mathbf{ab}} \;\leq\; T_{\mathbf{ab}}\big|_{\pi_i(\mathbf{a})=v(\mathbf{a}) \text{ for all } i}$$

with strict inequality whenever $\mathcal{R} \neq \emptyset$. By condition (4), the right-hand side equals $T^{\text{nd}}_{\mathbf{ab}}$. The analogous argument with reversed inequality applies to D-gaining transitions. Hence $T(\mathbf{a}, \cdot) \preceq_{st} T^{\text{nd}}(\mathbf{a}, \cdot)$.

Step 2: T^{nd} is invariant under the $C \leftrightarrow D$ label swap.

Let $\sigma : S \to S$ be the map that swaps every C and D. By conditions (3) and (4), with common payoff 0, the only features of $(\mathbf{a}, \mathbf{b})$ that enter $T^{\text{nd}}_{\mathbf{ab}}$ are the mutation matrix and the set $\mathcal{R}$. Under action-invariant mutation the mutation matrix is σ-invariant; and $|\mathcal{R}|$ is σ-invariant because the set of players currently playing $b_{I(\mathbf{a},\mathbf{b})}$ in $\mathbf{a}$ has the same size as the set of players currently playing $\sigma(b_{I(\mathbf{a},\mathbf{b})})$ in $\sigma(\mathbf{a})$. Hence $T^{\text{nd}}_{\mathbf{ab}} = T^{\text{nd}}_{\sigma(\mathbf{a})\sigma(\mathbf{b})}$ for every transition, so the stationary distribution of T^{nd} is σ-invariant, giving $p_C^{(T^{\text{nd}})} = p_D^{(T^{\text{nd}})} = \frac{1}{2}$ [31].

Step 3: T^{nd} is stochastically monotone.

Condition (5) provides, for every $\mathbf{a} \leq \mathbf{a}'$, a coupling $(\mathbf{X}, \mathbf{X}')$ on a common probability space with $\mathbf{X} \sim T^{\text{nd}}(\mathbf{a}, \cdot)$, $\mathbf{X}' \sim T^{\text{nd}}(\mathbf{a}', \cdot)$, and $\mathbf{X} \leq \mathbf{X}'$ almost surely. For any upper set $U \subseteq S$, $\{\mathbf{X} \in U\} \subseteq \{\mathbf{X}' \in U\}$, so $T^{\text{nd}}(\mathbf{a}, U) \leq T^{\text{nd}}(\mathbf{a}', U)$, giving $T^{\text{nd}}(\mathbf{a}, \cdot) \preceq_{st} T^{\text{nd}}(\mathbf{a}', \cdot)$.

Applying Lemma 1 to $P = T$ and $Q = T^{\text{nd}}$ (which satisfy (i) by Step 1 and (ii) by Step 3) yields $\pi^T \preceq_{st} \pi^{T^{\text{nd}}}$. Since n_C is monotone in the componentwise order,

$$p_C^{(T)} \;=\; \frac{1}{N}\,\mathbb{E}_T[n_C] \;\leq\; \frac{1}{N}\,\mathbb{E}_{T^{\text{nd}}}[n_C] \;=\; p_C^{(T^{\text{nd}})} \;=\; \frac{1}{2}.$$

□

This shows that even in a game where an action change $D \to C$ would result in an increase in payoff for a given player, under a purely extrinsic population dynamic we cannot achieve a probability of cooperation above that of the neutral drift baseline. Purely extrinsic processes always favour the pointwise dominant strategy, regardless of which strategy provides the Nash equilibrium for the one shot game. Thus, a player's chosen strategy does not necessarily favour the action type which maximises their payoff in *either* the one-step or the long-run strategy distribution, instead favouring the strategy which is pointwise dominant over its peers. We now show this result in the context of the public goods game with heterogeneous contributions.

3.4 Purely extrinsic population dynamics in the Public Goods Game

In Figure 3, we explore Theorem **??** numerically in the context of the public goods game. In panels (a) through (e), we can see this limit as a hard ceiling on the value of p_C which no parameter set in our data is able to exceed. This includes parameter combinations which have high values of $r > N$, and so we see that despite full cooperation being the Nash equilibrium of the one shot game, and increasing the payoff of players in the one-step distribution, the pointwise dominance of defection's fitness over that of cooperation leads to defection becoming the dominant strategy in a repeated game according to a purely extrinsic population dynamic. This not only leads to a worse average payoff for all players in the population in the steady state, but also means that when $r > N$, worsening moves occur with a higher probability than changes of action types which improve a player's payoff. In other words, even when a switch $D \to C$ would always improve a player's payoff, the player will still prefer to remain a defector under a purely extrinsic population dynamic, due to the greater fitness of a defector in a given state.

In Fermi imitation dynamics, we can also note the process' invariance under r. Figures 3a and 3c show an identical distribution of p_C for all values of r. The latter of these two figures also give that an increase in β will have a negative effect upon p_C, as the value $\Delta(\pi)$ will always be negative for a $C \to D$ transition, and positive for a $D \to C$ transition. As the process with $\beta = 0$ gives us neutral drift, we see our result in practice, as moving towards a less rational process increases p_C in all empirical cases, up to the neutral-drift baseline. p_C in the Moran process increases with the value of r, however once again we see that it has a hard limit at $p_C = \frac{1}{2}$, as a given defector is always more likely to be chosen for reproduction than a given cooperator. The Moran process' ability to consider the r parameter leads it to have a higher p_C than Fermi imitation dynamics in most parameter combinations, as shown in Figure 3f.

4 Intrinsic Population Dynamics

Another classification of population dynamics is intrinsic population dynamics. In such population dynamics, a player compares their payoff to some attribute or payoff other than the payoff of other players in the population. We shall see two such population dynamics - aspiration dynamics, which compares a player's payoff to their individual baseline aspiration, and introspection dynamics, which compares a player's payoff to their counterfactual payoff when they change action type.

4.1 Aspiration Dynamics

Aspiration dynamics models an update rule based on how far away a player's performance in a game is from an aspirational constant fitness value:

1. A player i is chosen with probability $\frac{1}{N}$ to reconsider their strategy $a_{i,k}$
2. This player changes strategy with a probability $\phi(\pi_i(\mathbf{a}) - A_i)$

Here, A_i is player i's *aspiration* - the payoff which they desire to make from a given game. A player will accept a new strategy with a higher than random probability if their current payoff is lower than their aspired payoff. This dynamic is almost always defined for two actions, traditionally cooperation and defection [7], however it has been defined for games with 3 or more actions [50]. In this paper, we consider a homogeneous aspiration value A, and the two-action formulation of aspiration dynamics as presented in [7].

The process transitions between states according to the transition matrix:

$$T_{\mathbf{ab}} = \begin{cases} \dfrac{1}{N}\,\phi(\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a}) - A_{I(\mathbf{a},\mathbf{b})}) \left(1 - \sum_{j=1}^{k} \mu_{I(\mathbf{a},\mathbf{b}),j}\right) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),\mathbf{b}_{I(\mathbf{a},\mathbf{b})}}}{N} & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}), \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b}. \end{cases} \tag{6}$$

The traditional two-action model of aspiration dynamics is very specific in which games it can be applied to. When aspiration is considered for three or more action types [50], a player chooses which action to change to with equal probability. Thus, players in 3-action games do not choose their strategy in the next state based on the fitness of said

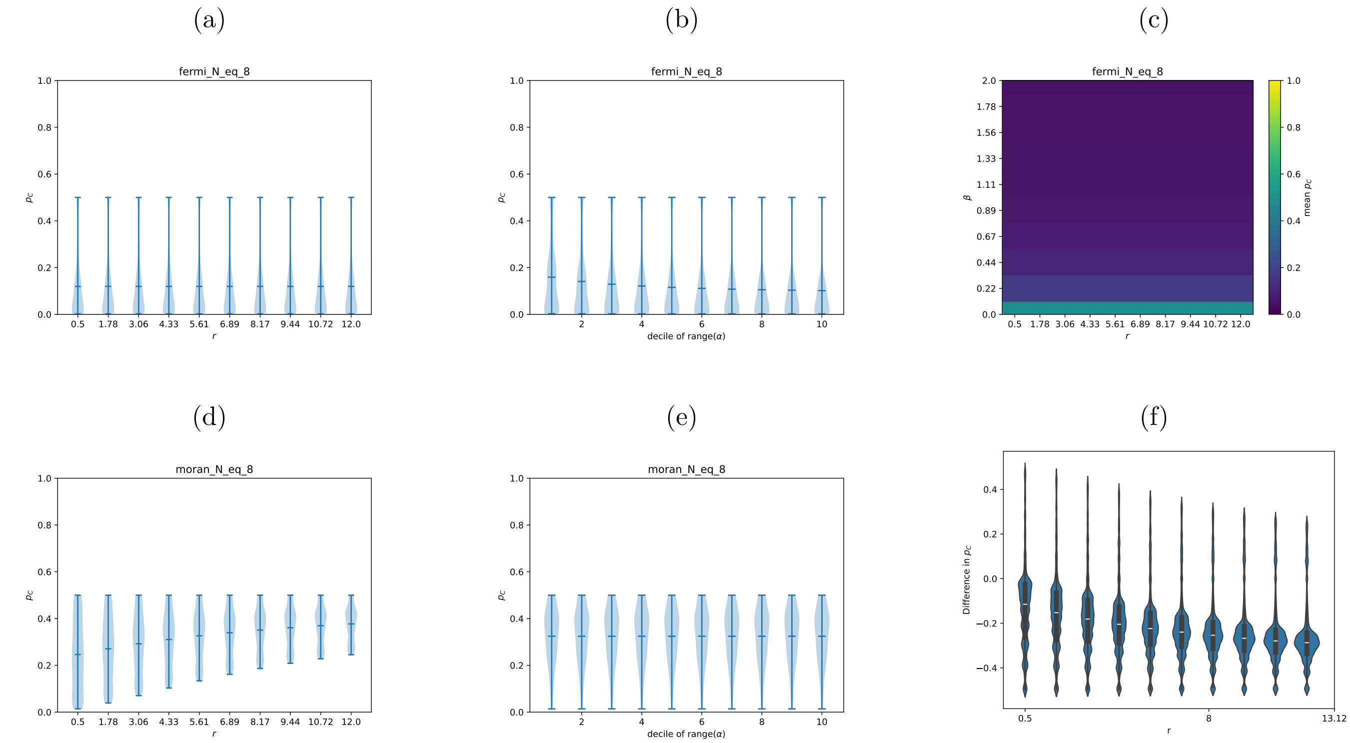


Figure 3: (a): The distribution of p_C in Fermi imitation dynamics for different values of r. Since Fermi imitation dynamics is invariant under the value of r, this distribution does not change, and for all values of r it admits a maximum value of p_C below the neutral-drift baseline $p_C = \frac{1}{2}$.
(b): The distribution of p_C for Fermi imitation dynamics for different levels of heterogeneity in contribution. The greater the decile of range(α_i), the greater the heterogeneity in contribution.
(c): A heatmap of mean(p_C) for values of r and β for Fermi imitation dynamics. We see that there is a negative correlation between β and p_C.
(d), (e): The equivalent of (a) and (b), respectively, for the Moran process
(f): The distribution of difference in p_C between Fermi imitation dynamics and the Moran process for different combinations of shared parameters.

strategy, but instead update according to "trial and error" in order to reach their aspiration, and thus players may accept worsening moves with a high probability.

This population dynamic spends a large amount of time in states where most players achieve their aspired payoffs. In this case, players will have a lower probability of switching strategy, compared to a state where players are underperforming in comparison to their target A. Thus, cooperation emerges more frequently in aspiration dynamics under larger values of A, as more cooperators are required in order for all players to reach their aspired payoff. At a lower value of A, the reverse is true - fewer cooperators are required for all players to reach their target payoff.

4.2 Introspection Dynamics

In introspection dynamics [5, 6], individuals are selected to consider if a change of their action choice would lead to an increase in payoff. The new action is selected with a probability that depends on the potential increase and the selection intensity β:

1. A player i is chosen with probability $\frac{1}{N}$ to reconsider their strategy $a_{i,k}$
2. Player i randomly chooses another possible strategy, a_{il}, from the set of all possible strategies.
3. The chosen player replaces their strategy with probability $\phi(\Delta\pi_{I(\mathbf{a},\mathbf{b})})$, where $\Delta\pi_{I(\mathbf{a},\mathbf{b})} = \pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a}) - \pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{b})$ is the difference between the fitness of the player's current strategy and the possible fitness of the new strategy.

This process is well suited to a heterogeneous population. This is because in some cases using imitation dynamics, a player may copy a strategy due to its fitness when played by another individual in the population, despite the fact that such a strategy actually has a lower payoff for the focal player than their current strategy. In introspection dynamics, players do not consider the fitness of other individuals, and so cannot be fooled by high fitness strategies with attributes different to that of the focal player.

The process transitions between states according to:

$$T_{\mathbf{ab}} = \begin{cases} \dfrac{1}{N(k-1)}\,\phi(\Delta\pi_{I(\mathbf{a},\mathbf{b})})\left(1-\sum_{j=1}^{k}\mu_{I(\mathbf{a},\mathbf{b}),j}\right) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),\mathbf{b}_{I(\mathbf{a},\mathbf{b})}}}{N} & \text{if } \mathbf{b} \in \mathrm{Neb}(\mathbf{a}), \\ 0 & \text{if } \mathbf{b} \notin \mathrm{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b}\in S\setminus\{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b}. \end{cases} \tag{7}$$

where k is the number of actions available to each individual. It has been shown [**?**] that in state-independent games, such as the public goods game and certain forms of the prisoners dilemma, introspection dynamics can be modelled as N individual Markov chains, where each player changes action independently of the action types of the rest of the state. Thus, p_C breaks down into the mean p_i, the probability that player i cooperates. This makes introspection dynamics an unrealistic model of the evolution of cooperation in many common abstractions of real-world interactions, as players do not consider the current success of a strategy when choosing to consider or accept it, but only consider the fitness of a strategy in the next step after adoption.

Introspection dynamics can also be seen to follow a similar logic to classical game theory, where a well informed and rational actor maximises their own payoff. We can contrast this with purely extrinsic population dynamics, which are the traditional method of modelling populations in evolutionary games, where actors are not typically well-informed and choose their next strategy according to fitnesses in the current population, without knowledge of the impact of their transition on their fitness. Introspection dynamics allows a population to evolve by choosing new action types with full knowledge of the impact of their decision in the following one-shot game when playing a repeated game. We shall now see the impact which this has on the emergence of cooperation in public goods game.

4.3 Intrinsic population dynamics in a public goods game

In Figure 4, we see that intrinsic population dynamics are able to exceed the ceiling on p_C which is imposed on purely extrinsic population dynamics. In the case of aspiration dynamics, p_C does not reach excessively low values as in the other processes we have discussed. This is because for all players to make at least some amount of profit, at least one player must be cooperating at all times. If no players cooperate, then no player will reach their target payoff threshold. Although the ceiling is not exceeded if $r < N$ (both shown in Figure 4a and noted in [7]), we must still have multiple players cooperate in order to allow all players to reach the cooperation threshold, and the furthest above the aspiration threshold is reached when many players cooperate (although in some levels of heterogeneity, it may be that a certain number of players will

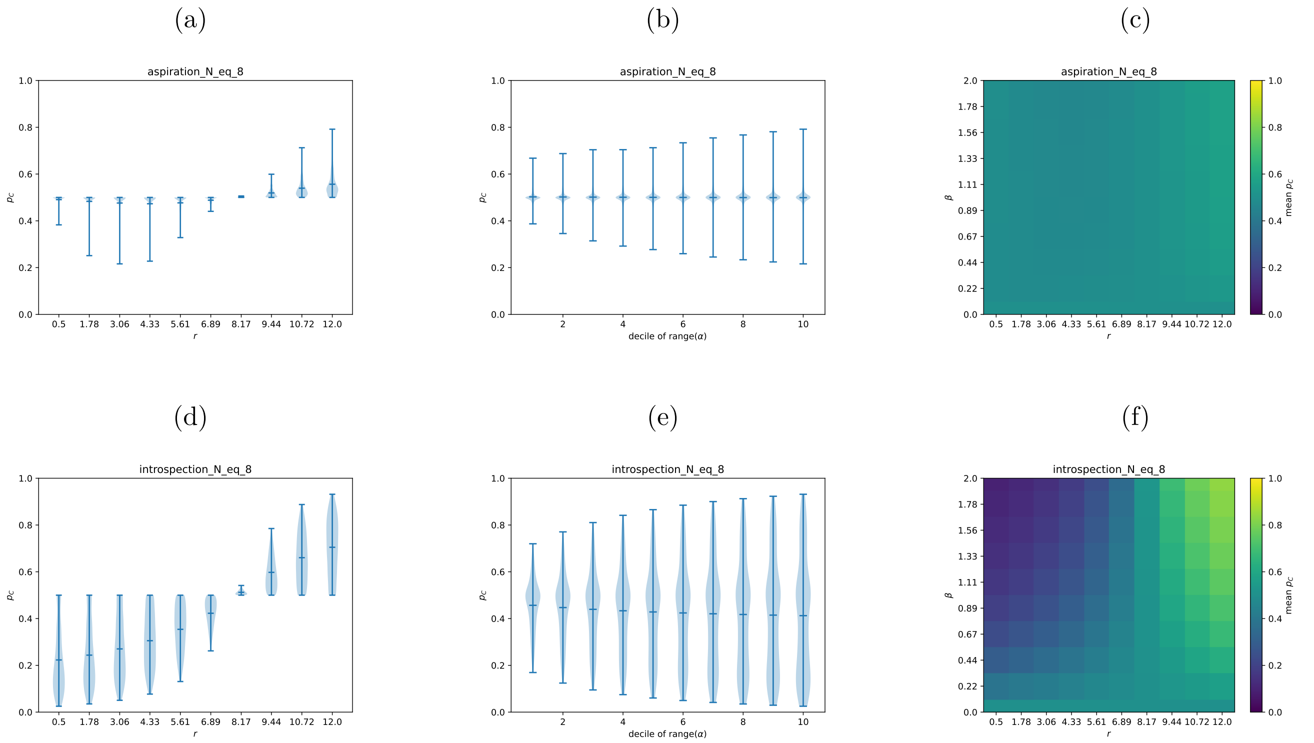


Figure 4: (a): The distribution of p_C in aspiration dynamics. For medium values of r, players are able to achieve their aspired fitness with fewer cooperators, and so the value of p_C is reduced. However, when $r > N$, players achieve a higher payoff in a state $\mathbf{a}$ where they cooperate than in the state $\mathbf{a}_{-i}$ where they do not cooperate. Thus we have $p_C > \frac{1}{2}$ in this case
(b): The distribution of p_C for aspiration dynamics for different levels of heterogeneity in contribution. The greater the decile of range(α_i), the greater the heterogeneity in contribution.
(c): A heatmap of mean(p_C) for values of r and β for aspiration dynamics. We see that there is a negative correlation between β and p_C. The mean p_C of aspiration dynamics is usually in the region (0.4, 0.6)
(d),(e), (f): The equivalent of (a), (b), and (c), respectively, for introspection dynamics

defect more often in the steady state due to the large contributions needed to be made). The ceiling is exceeded at the threshold $r > N$, as for all players, $\pi_i(\mathbf{a}) - A$ is greater when cooperating than defecting in a given state $\mathbf{a}_{-i}$, and thus the transition $D \to C$ will always occur with a higher probability than the reverse. For intermediate values of r, aspiration dynamics can achieve its lowest p_C, as with a low value of A, players are able to achieve their aspiration by free-riding in a state with fewer defectors. Figure 4c shows that the mean value of p_C remains fairly steady for the values of r and β. The aspiration level is the biggest indicator of cooperation in the population in a linear public goods game - it has been shown previously that for a high aspiration level, cooperation emerges at a level near the neutral-drift baseline, but for a low aspiration level defectors dominate the population [7].

In the case of introspection dynamics, we can see that the ceiling is once again breached at exactly the threshold $r = N$.

Proposition 1 ($r = N$ threshold)**.** *For any heterogeneous population with $\alpha_i > 0$ for all i, the introspective evaluation step of Introspection Dynamics favours cooperation if and only if $r > N$.*

Proof. Individual i evaluates switching from defection to cooperation by comparing their payoff $\pi_i(\mathbf{a})$ in the current state to the payoff $\pi_i(\mathbf{b})$ they would receive in the state where they cooperate and all others remain unchanged. Using the convention $\Delta\pi_i = \pi_i(\mathbf{a}) - \pi_i(\mathbf{b})$,

$$\Delta\pi_i = \alpha_i - \frac{r\alpha_i}{N} = \alpha_i\Big(1 - \frac{r}{N}\Big).$$

Since $\alpha_i > 0$, this is negative if and only if $r > N$. Since ϕ is a decreasing function, $\phi(\Delta\pi_i) > \frac{1}{2}$ precisely when $\Delta\pi_i < 0$, i.e. when $r > N$, and $\phi(\Delta\pi_i) < \frac{1}{2}$ when $r < N$. □

When $r < N$, the introspective step will always favour defection. This behaviour gives us a system where under introspection dynamics, players are able to choose the strategy which will maximise their own fitness at any given time step. The ability for this process to recognise the impact of their transition leads to a realistic model of real-world decision making, and a suitable population dynamic for a heterogeneous population. However, as introspection dynamics can become independent of the rest of the population in a state-independent game, it cannot always fully model learning in a real world context without making assumptions which may not reflect how individuals make decisions in the real world. Figure 4f shows this further - a stronger choice intensity favours cooperation if a switch $C \to D$ favours cooperation, and favours defection if the reverse is true. While this is the clear rational behaviour, it is not true for Fermi imitation dynamics, as shown in Figure 3c. The intrinsic step can therefore be said to favour the strategy which admits the Nash equilibrium in the public goods game.

5 Introspective Imitation Dynamics

We introduce our novel population dynamic *introspective imitation dynamics*. This population dynamic combines the introspective step of introspection dynamics with the fitness-proportional selection of the Moran process, in order that each individual takes into account both the fitness of strategies in the current state, and the counterfactual payoff obtained when accepting a new strategy. This models both extrinsic learning and intrinsic learning. We define the introspective imitation process as follows:

1. A focal player is chosen at random to reconsider their strategy
2. Another player a_j in the population is chosen with a probability proportional to their fitness within the population to have their strategy considered, as in the Moran process.
3. The focal player changes their strategy with probability $\phi(\Delta\pi_i)$, as in introspection dynamics

Recall that $f_i(\mathbf{a}) = 1 + \epsilon \cdot \pi_i(\mathbf{a})$. We define the transition matrix T of an introspective imitation process as follows:

$$T_{\mathbf{ab}} = \begin{cases} \dfrac{1}{N}\dfrac{\sum_{j:\, a_j = b_{I(\mathbf{a},\mathbf{b})}} f_j(\mathbf{a})}{\sum_l f_l(\mathbf{a})}\, \phi(\Delta\pi_{I(\mathbf{a},\mathbf{b})})\left(1 - \sum_{j=1}^{k} \mu_{I(\mathbf{a},\mathbf{b}),j}\right) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),\mathbf{b}_{I(\mathbf{a},\mathbf{b})}}}{N} & \text{if } \mathbf{b} \in \mathrm{Neb}(\mathbf{a}), \\ 0 & \text{if } \mathbf{b} \notin \mathrm{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S\setminus\{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b}. \end{cases} \quad (8)$$

This models a behaviour where individuals learn from each other, but do not blindly trust that the strategy will work for themselves solely because it worked for another player in the population. Instead, players look to the fitness of others in the population to see what action they *could* play, but judge based on their own fitness to decide what action they *should* play. This means that introspective imitation dynamics is well-suited to a heterogeneous population, but does not ignore the success of strategies in the rest of the population.

We see that this dynamic does not break down into individual Markov chains like introspection dynamics due to the extrinsic step, however it is shown in Figure 5b able to exceed the extrinsic population ceiling under the right conditions due to the intrinsic step. It is also defined for any number of strategies, and will consider each strategy proportional to both its fitness within the state and the counterfactual payoff which the focal player is able to obtain by adopting the new action type. Finally, as the value of $\Delta(\pi)$ is cross-state as in introspection dynamics, introspective imitation dynamics does not cancel out certain parameters in all comparisons, as in Fermi imitation dynamics. We can thus say that all limitations discussed for population dynamics in this paper are not present in the formulation of introspective imitation dynamics.

Introspective imitation dynamics also has the advantage that it considers both a choice intensity β and a selection intensity ϵ. This means that the parameters ϵ and β are both tunable to emphasise either the extrinsic or intrinsic step of the dynamic. This allows for the dynamic to be adjusted to suit real-world modelling, depending on whether a certain individual may choose actions in order to improve their payoff in comparison to others, or in order to maximise their own personal gain.

5.1 Introspective Imitation Dynamics in the Public Goods Game

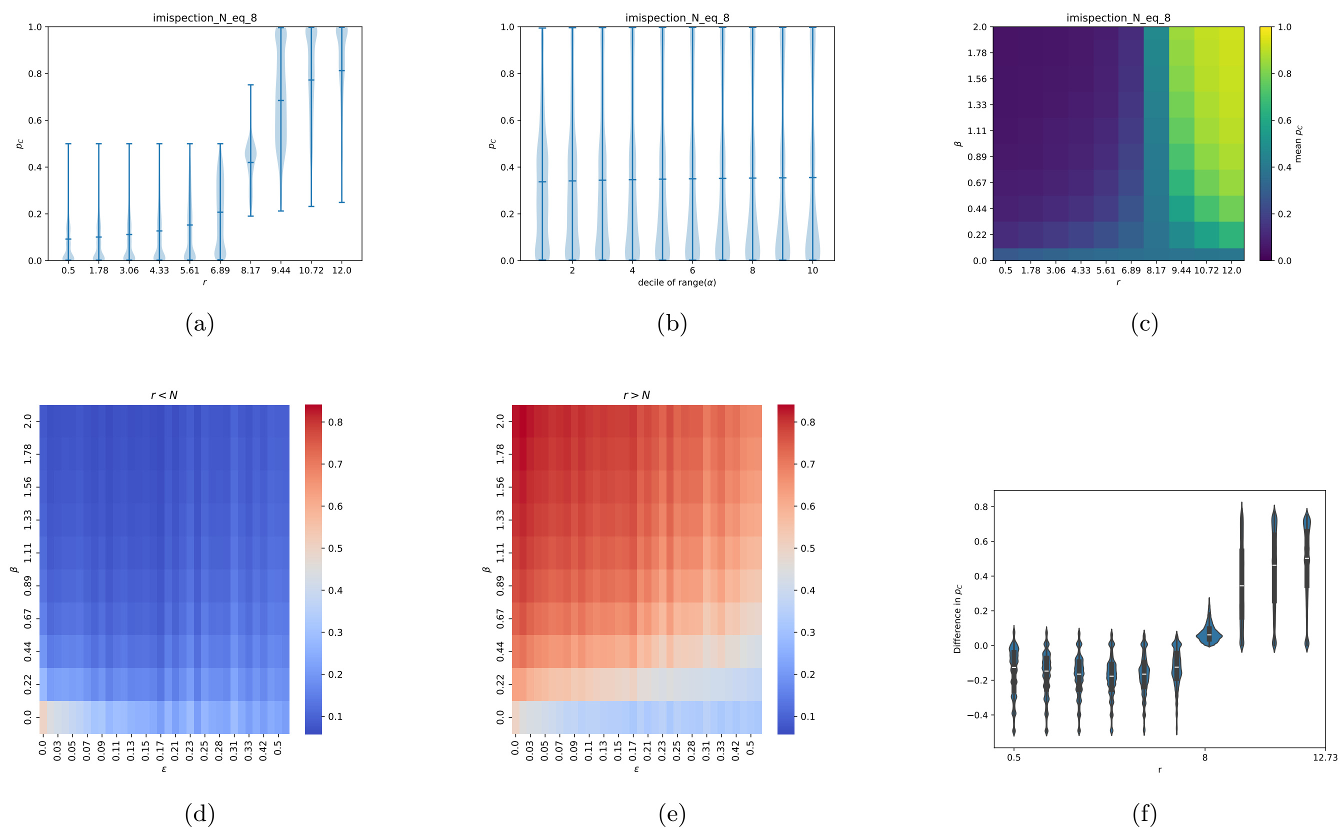


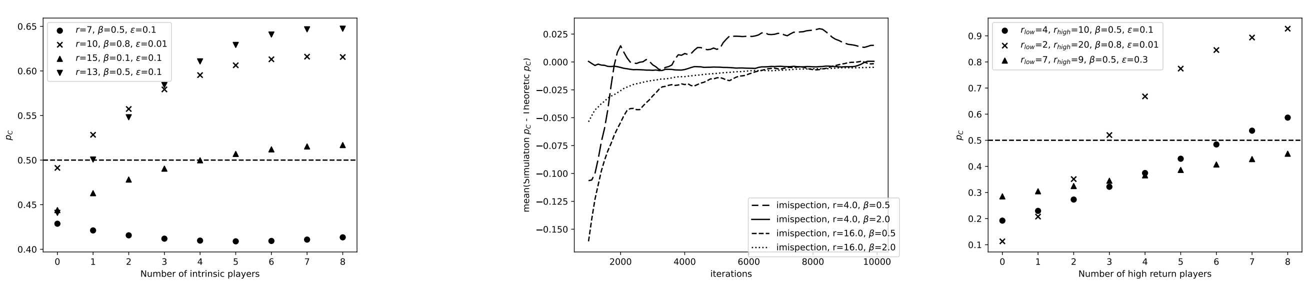
Figure 5: (a): the distribution of p_C in introspective imitation dynamics for various values of r.
(b): the distribution of p_C in introspective imitation dynamics for various levels of heterogeneity. The greater the value of range(α), the greater the contribution heterogeneity.
(c): A heatmap of mean(p_C) in introspective imitation dynamics for varying r and β. We see a threshold for the effect of a stronger choice intensity at $r = N$.
(d),(e): Heatmaps of the effect of both selection and choice intensity on mean(p_C) for $r > N$ and $r < N$.
(f): The difference in p_C between introspective imitation dynamics and the Moran process for populations with shared parameters, specifically for different values of r.

We see in Figure 5a that in the same manner as the purely intrinsic population dynamics, introspective imitation dynamics is able to exceed the $p_C = \frac{1}{2}$ ceiling which is imposed by purely extrinsic population dynamics. Despite the extrinsic step in the dynamic, the intrinsic step is sufficient to enable the dynamic to admit high values of p_C under the right parameters. We can see in Figure 5a that this occurs exactly at the threshold $r = N$: the same threshold that we see $\phi(\Delta(\pi))$ begin to favour cooperation in introspection dynamics. Before this threshold, we actually see in Figure 5f that introspective imitation dynamics favours defection more severely than in the standard Moran process, as both steps supress the probability of a $D \to C$ transition. We can see, therefore, that the intrinsic step of introspective imitation dynamics, as in introspection dynamics, will favour the strategy which admits a nash equilibrium in the one-shot public goods game (i.e, cooperation if the Nash equilibrium is for all players to cooperate, and defection if it is for all players to defect).

This does not necessarily imply that introspective imitation dynamics admits a value of p_C which favours cooperation in line with ϕ when $r > N$, as certain choices of ϵ and β may cause the process to emphasise the extrinsic step, which will always favour defection, over the intrinsic, as seen in Figure 5e. We have seen previously for introspection dynamics that a stronger choice intensity in the intrinsic step will favour cooperation if $r > N$ and favours defection if the inequality is reversed - this remains true in introspective imitation dynamics. A stronger selection intensity on the other hand always favours defection. There is no result which says, for a given pair of choice and selection intensity that favours defection, that there is no value of r which admits a $p_C > \frac{1}{2}$ - in fact it is likely that such a value exists outside of the range of our empirical data, as the difference between a cooperator and defector in the extrinsic step becomes minimal, and the intrinsic step strongly favours cooperation.

Previously, we saw that heterogeneity had little effect on p_C in the Moran process (Figure 3e), and that a greater level of heterogeneity caused a lower mean and more polarised extreme values in introspection dynamics (Figure 4). In introspective imitation dynamics, heterogeneity causes the mass of the distribution to approach the extremes of p_C. This effect can once again be attributed to the threshold $r = N$. One of two systems then appear. If $r < N$, high contributors are tempted with a large improvement in payoff if they defect as $\alpha_i \left(\frac{r}{N} - 1\right)$ is much lower, and much lower contributors are both selected more often than their medium-contributing counterparts in a population with low heterogeneity. Thus, high contributors cooperate much less in a more heterogeneous population with a low r. If $r > N$, then the return $\frac{r\alpha_i}{N}$ given by a high contribution is much larger in a more heterogeneous population. Thus, the intrinsic step of the population dynamic will favour cooperation much more strongly in a population with a greater level of heterogeneity if $r > N$. We see this polarisation once again favours the strategy which maximises the overall payoff of a given player in the one-shot game.

6 Conclusion



(a) (b) (c)

Figure 6: (a): p_C in various parameter combinations at $N = 8$, varying the number of players using an intrinsic population dynamic. 0 intrinsic players gives the p_C of the Moran process, whereas 8 gives us the p_C in introspective imitation dynamics. We take a linear population with a maximum contribution of 8.
(b): for various parameter combinations, the mean value over numerous simulations of an 8 player population playing introspective imitation dynamics converging to the theoretic p_C.
(c): p_C for a heterogeneous r in introspective imitation dynamics. 0 high return players corresponds to a population with homogeneous return r_{low}. 8 high return players corresponds to a population with a homogeneous return r_{high}.

We have introduced and analysed a framework for heterogeneous evolutionary games under five distinct population dynamics in three categories, as shown in Figure 1. We have defined the categorisation of a *purely extrinsic population dynamic*, and given rigorous formulations of two widely used examples in the Moran process and Fermi imitation dynamics. We have shown that dynamics under this classification have a hard ceiling on the probability of cooperation p_C in social dilemmas where a defector *always* outperforms a cooperator in a given state. In such a game, purely extrinsic population dynamics are unable to exceed the neutral-drift baseline. We then give formulations of two intrinsic population dynamics - aspiration dynamics in which players change strategy based on their performance in relation to a target payoff, and introspection dynamics in which players compare their current payoff with their counterfactual payoff which they could obtain by switching action type. We have also shown empirical results for the public goods game with heterogeneous contributions which confirm the ability for intrinsic population dynamics to exceed the population ceiling imposed by purely extrinsic processes. Finally, we have introduced a novel population dynamic in Introspective imitation dynamics with both an extrinsic and intrinsic step. Our empirical results confirm that this dynamic is able to overcome the ceiling of purely extrinsic population dynamics, but still allows players to take into account the performance of strategies within the current state. We establish the threshold $r = N$ for which the intrinsic step in introspection dynamics and introspective imitation dynamics are able to overcome the neutral-drift baseline.

Further work on this may look further into the effect of hybrid population dynamics, similar to the combination of Moran and Fermi processes in [25], where extrinsic and intrinsic population dynamics are played in a mixed population. Figure 6a shows three populations where the population dynamic is treated as a heterogeneous attribute, with players learning according to either the Moran process or introspective imitation dynamics. We see that under different parameters, a different number of players may possess an intrinsic learning method in order to exceed the neutral drift baseline p_C. Figure 6b shows that simulations of introspective imitation dynamics converge to the theoretic value of p_C across a large number of iterations. Further study may make use of mixing times [24] in order to show which population dynamics converge to their theoretic p_C in the fewest iterations. Finally, the heterogeneity considered in this paper is in the player's contributions, however the threshold for introspective imitation dynamics occurs at the point $r = N$. Figure 6c shows that in introspective imitation dynamics with certain heterogeneous values of r_i, fewer players are required to receive a return $r > N$ in order to exceed the cooperation threshold. Additionally, we see in both Figures 6c and 5e that under certain values of ϵ and β, even a population which is entirely contributing a high value of r will be unable to exceed the p_C threshold, as the extrinsic step of the process occurs with a higher rationality. The relationship between such values and the heterogeneity of r may be a focus of further study.

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